

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Process Control & Safeguarding Philosophy

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R01	02-07-2022	Issued for Review
R02	17-12-2022	Re-Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PSE-PHY-002

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEM as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

The document, in consideration of relational design elements & concepts, forms the technical basis for the development of the facilities Project Specifications & tendering of contracts for the project's implementation.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
AASHTO	American Association of state Highway Transportation Officials
AGEF	Above Ground End Facilities
ALARP	As Low as Reasonably Practicable
API	American Petroleum Institute
ASTM	American Society of Testing and Materials
BOD	Basis Of Design
CP	Cathodic Protection
CPT	Cone Penetrometer Test
CTMS	Custody Transfer Metering Station
DB&B	Double Block and Bleed
DCQ	Daily Contractual Quantity
CSC	Car Seal Closed
CSO	Car Seal Open
DBB	Double Block and Bleed
DEP	Design and Engineering Practice
DPR	Department of Petroleum Resources
EBD	Emergency Blowdown
ESD	Emergency Shutdown
ESDV	Emergency Shutdown Valve
F&G	Fire and Gas
FEED	Front End Engineering Design
FGS	Fire, Gas and Smoke detection systems
GPP	Gas Processing Plant
GTG	Gas Turbine Generators
HAZID	Hazard Identification
HAZOP	Hazard Operability
HEMP	Hazards and Effects Management Process

HFE	Human Factors Engineering
HMI	Human Machine Interference
HPU	Hydraulic Power Unit
HRA	Health Risk Assessment
HSES	Health, Safety, Environment and Security
HSSE & SP	Health, Safety, Security, Environment and Social Performance
ICSS	Integrated Control and Safety System
IEC	International Electro-technical Commission
ISPM	Industry Standard Process Modules
JV	Joint-Venture
LC	Locked Closed
LCD	Liquid Crystal Display
LO	Locked Open
LOI	Local Operator Interface
LOS	Line of Sight
LRL	Lower Range Limit
LRV	Lower Range Value
LIRA	Logistics and Infrastructure Resource Assessment
MAC	Main Automation Contractor
MOS	Maintenance Override Switch
NAG	Non-Associated Gas
OPC	OLE for Process Control
OPPS	Overpressure Protection System
PCD	Process Control Domain
PDMS	Plant Design and Management System
PEPF	Package Engineering, Procurement, and Fabrication
PIG	Pipeline Inspection Gauge
PLC	Programmable Logic Controller
PSV	Pressure Safety Valve

PVC	Polyvinyl Chloride
RTR	Reliability Test Run
SCE	Safety critical elements
SDV	Shutdown Valve
SI	International System Units
SIL	Safety Integrity Level
SRB	Sulphate Reducing Bacteria
SIS	Safety Instrumented System
SSD	Station Shutdown
SVI	Single Valve Isolation
SWB	Steel Wire Braiding
URV	Upper Range Value
USC	US Customary Units
XLPE	Cross Linked Polyethylene
TEG	Tri-ethylene Glycol
TQ	Top Quartile

1. Introduction

1.1. Background Information

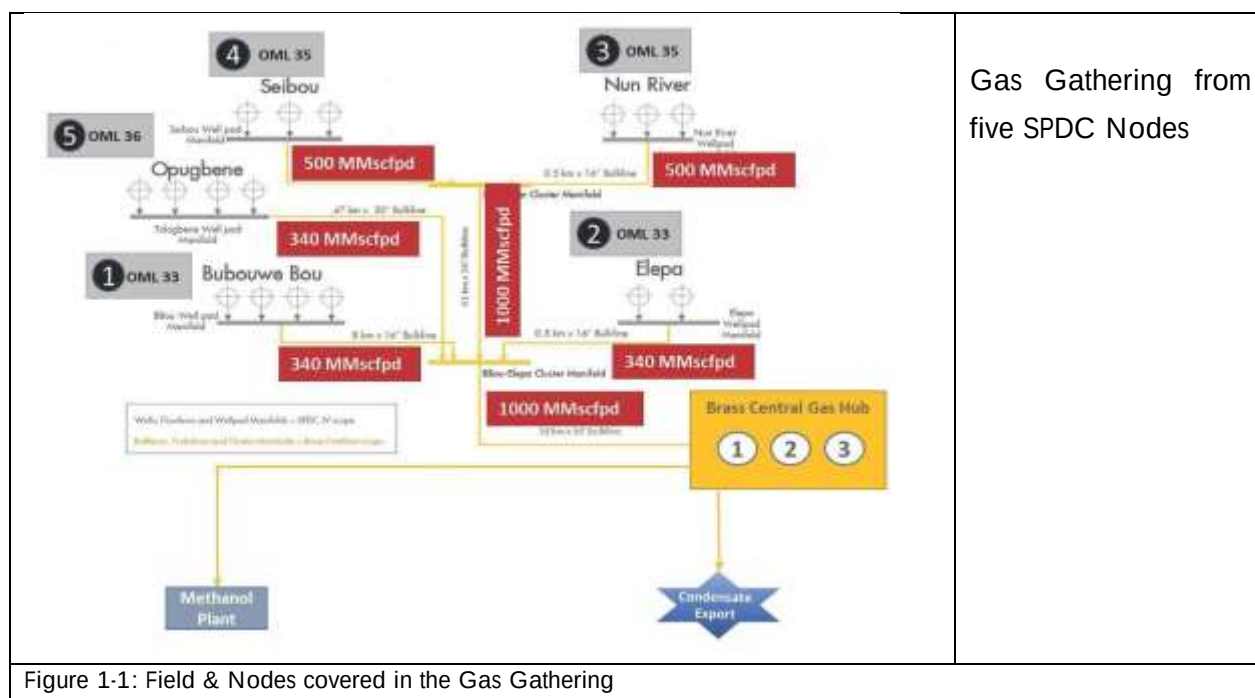
Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a greenfield methanol production facility in Akabel (Odioma), Brass LGA Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields, namely: Bubouwe Bou & Elepa, in OML 33, Nun River, Seibou, in OML 35, and Opugbene in OML 36.

The Project involves the development, construction, and operation of a 10,000 metric tonnes per day (MTPD) capacity methanol production plant comprising the following elements:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel, - as shown in Figures 1.1, 1.2, & 1.3.
- Product Export Lines and associated Offshore Single Buoy Mooring (SBM) for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) which are required for the operation of the processing plants.

Tata Consulting Engineers Limited (TCE) is the Project Management Consultant for the project.

The Nodal Gas Gathering Schematics



The Nodal Gas Gathering Schematics (contd.)

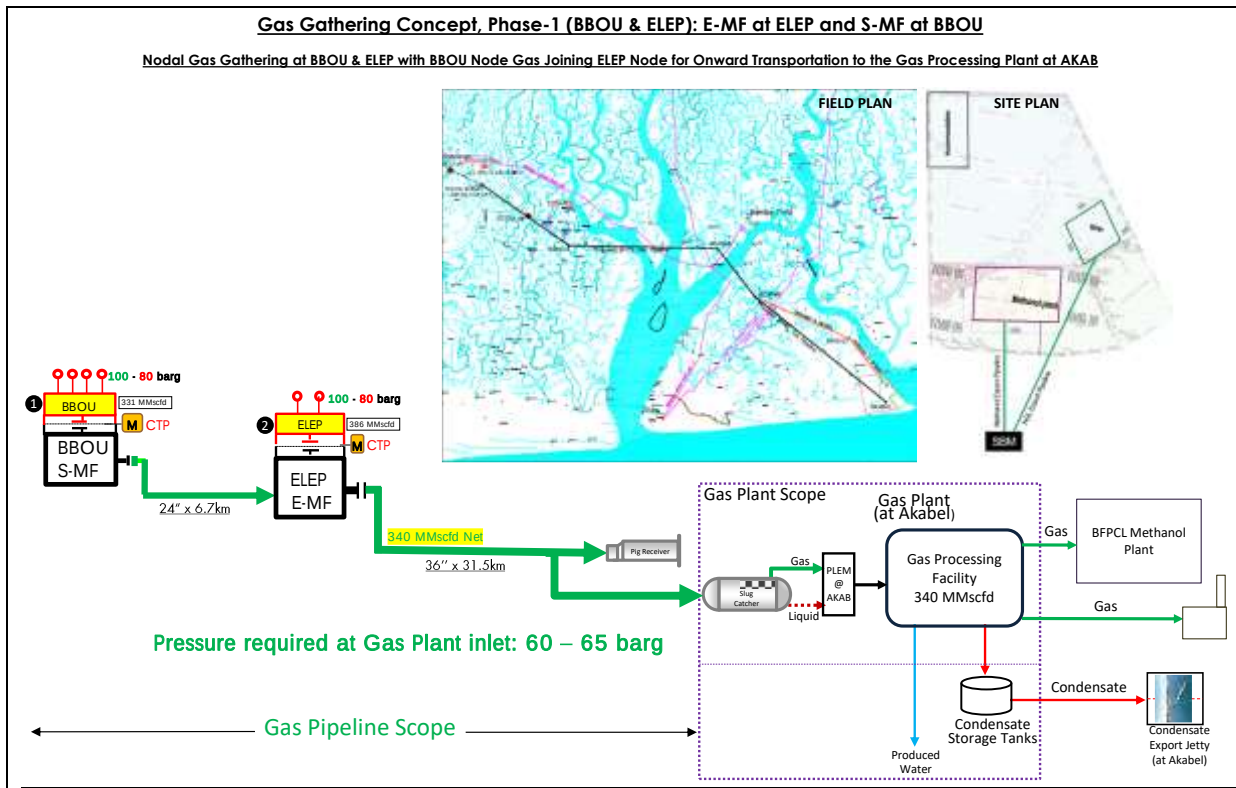


Figure 1-2: Gas Gathering Concept, Phase 1

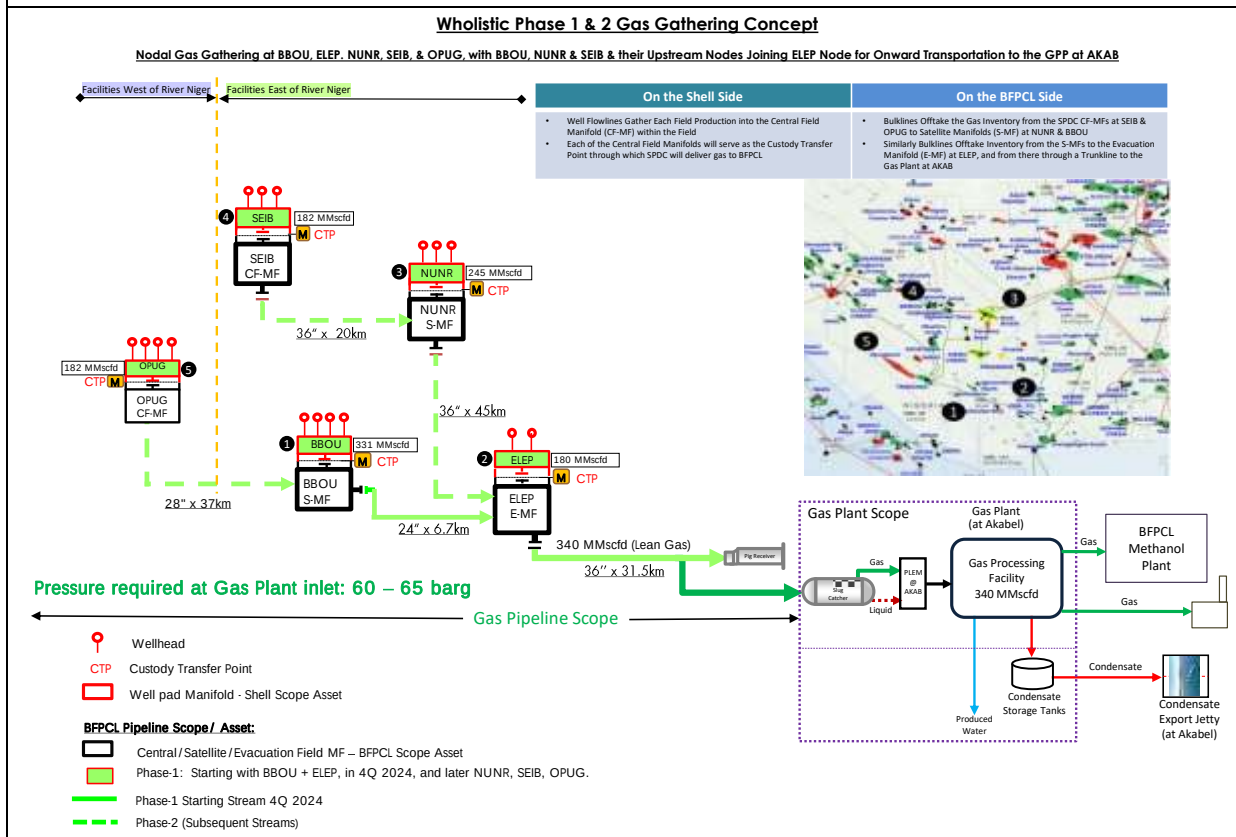


Figure 1-3: Wholistic Gas Gathering Concept, Phase 1 & 2

1.2. Purpose of Document

This purpose of this document is to describe the safeguarding philosophy for facilities in the FEED for Gas Gathering Pipelines & Associated Infrastructure Project.

2. Regulations, Codes And Standards

2.1. Concept Engineering Referenced Documents

Document Number	Title
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA/SPDC-BFPC UGS-GEP/SVY-RPT-001	Pipeline Route Selection Study
EA/SPDC-BFPC UGS-GEP/PRC-RPT-001- 3	Process Simulation Study – HYSYS, PIPESIM, & OLGA
EA/SPDC-BFPC UGS-GEP/MAT-RPT-001	Pipeline Materials Selection & Corrosion Analysis Study
EA/SPDC-BFPC UGS-GEP/ELE-RPT-001	Manifold Platforms Power Supply Options
EA/SPDC-BFPC UGS-GEP/MEC-RPT-001	Manifold Platforms Concept & Integration with SPDC
EA/SPDC-BFPC UGS-GEP/PRC-RPT-002	Gas Processing Technology Option
6064/PLRS/001a-PRP	Onshore Pipeline Route Survey for BFCL; Topographical Surveys
6064/PLRS/001PRP	Onshore Pipeline Route Survey for BFCL; Soil Investigation Surveys
40km Gas Supply Pipeline Project ESIA	Environmental and Social Impact Assessment Report prepared by Messrs. Fundamental Integrated Site Appraisal Services Limited - Draft Report November 2020.

2.2. Shell Design and Engineering Practices (DEPs)

Applicable DEPs, supplements and MESC SPE Codes are as referenced in various discipline dossiers. The design shall be based on DEP Feb 2019 (or latest, as advised by the EMPLOYER) some of which are listed below.

Document Number	Document Title
DEP 31.40.00.10-Gen.	Pipeline Engineering
DEP 31.40.10.13-Gen	Design of Pipeline Pig Trap Systems
DEP 31.40.10.14-Gen	Pipeline Overpressure Protection
DEP 31.40.21.33-Gen.	Pig signallers: Intrusive type
DEP 31.40.30.30-Gen	Concrete coating of line pipe
DEP 31.40.30.31-Gen	External polyethylene and polypropylene coating for line pipe
DEP 31.40.40.38-Gen	Hydrostatic pressure testing of new pipelines
DEP 31.40.50.30-Gen	Pre-commissioning of pipelines

Document Number	Document Title
DEP 31.40.60.11-Gen.	Pipeline leak detection

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Overpressure Protection Systems

The Pipelines emanate from manifolds at BBOU and Elepa. Facilities installed at each manifold location must be protected against overpressure as they tie-in to SPDC facilities with higher design pressure. In addition to mechanical protective devices (see section 3.1), protective instrumentation system shall be provided.

3.1. Pressure Safety Valves

Pressure safety valves used for protection against overpressure from SPDC facilities (upstream of custody transfer point) shall be selected according to standard size range provided by API STD 526.

3.1.1. Set Pressure

Pressure safety valve set pressures shall meet the requirements of the ASME Boiler and Pressure Vessel Code Section VIII, Pressure Vessels, Division 1 para UG-125, ASME B31.8 section 845 and API STD 520 Part I, Section 1.2.3.3 for all process and utility equipment.

The set pressure shall not exceed the design pressure of the system to be protected.

3.1.2. Allowable Overpressure

The overpressure allowed by Section VIII of the ASME Code is 21 percent of set pressure for a fire scenario and 10 percent for all operational failure scenarios for single pressure safety valve systems and 16 percent for multiple pressure safety valve systems.

3.1.3. Type

Pressure safety valves shall be conventional spring-loaded valves. The maximum back pressure for conventional valves shall be limited to 10 percent of set pressure.

3.1.4. Sizing and Selection

Pressure safety valves shall be sized for the governing case, using a realistic maximum developed back pressure. Both superimposed and built-up back pressures shall be established. The superimposed back pressure for valves discharging to the vent system shall be in the form of a range, the maximum of which reflects the vent system pressure when all other co-incident reliefs are active.

All sizing calculations shall subsequently be updated once actual system back pressures have been

calculated for governing overpressure scenarios when pressure safety valve Vendor data are available.

The sizing of pressure safety valves shall be based on the method contained in API STD 520 Part 1 Section 3.6.

3.1.5. Sparing

Pressure safety valves installed at BBOU and Elepa on the respective manifold header consist of one duty and one spare pressure safety valve (2x100%). All duty and spare pressure safety valves shall be identical including set pressure.

3.1.6. Installation

Discharge piping shall be free-draining, without any low point where liquids might collect.

Any isolation valves installed at safety valve inlet or at discharge piping shall be full-bore and suitably locked to ensure that the required relief capacity is always available.

Pressure safety valves shall be installed in the vertical position as close as possible to the equipment or system which is protected.

Pressure safety valves shall be installed in an accessible area and if required service platforms shall be provided for ease of maintenance.

3.1.7. Inlet Lines

To minimise the possibility of chattering, the inlet line to pressure safety valve shall be as short as it is practicable and sized to limit the total non-recoverable pressure loss between the protected section and the safety valve to 3% of the set pressure (as recommended by API RP 520). The pressure drop shall take account of the piping entrance loss and be based on the rated flow through pressure safety valve and pipe.

The inlet line to a pressure relief valve located in a gas system shall be free-draining to the process.

3.1.8. Discharge Lines

The discharge lines shall be sized to fulfill requirements presented in paragraph 3.1.6 above. Velocity in discharge piping shall not exceed 0.7 Mach number.

3.2. Overpressure Scenarios

The overpressure relieving devices protecting a pressure system shall be sized to provide adequate capacity for all overpressure scenarios.

Overpressure scenarios shall be evaluated from a detailed line-by-line review of the P&IDs to identify all potential sources of overpressure. Overpressure protection systems shall be sized to provide adequate capacity for all overpressure scenarios. This review shall also consider start-up, shutdown and intermittent plant operations, together with any pressure-related items from the HAZOP. Double jeopardy scenarios shall not be considered.

3.2.1. Closed Outlets

In case of closed outlets, there is a potential risk of overpressure of the Manifold piping and equipment as design pressure is lower than flowing tubing pressure of SPDC facilities upstream tie-in. Therefore, pressure safety valves shall be installed to protect the Manifolds and associated facilities against overpressure. Pressure safety valves shall be sized for maximum flow rate entering the manifold header at each of BBOU and Elepa.

Additionally, primary system protection shall be provided by protective instrumentation system.

The protection system shall lead to the following events:

- Alarm at high pressure.
- Emergency shutdown of the EH Pipeline inlet valve at GRF at high-high pressure.
- Relieving of gas through the PSV valve in case of overpressure.

3.2.2. Inadvertent High Pressure Supply from SPDC

Inadvertent supply pressure from SPDC facilities which is higher than the design pressure of the Manifolds and its facilities can cause a potential risk of over-pressurizing the System. Measures as described in paragraph 3.2.1 above will also be used as mitigating for this scenario.

Over-pressurizing vent piping system from inadvertent high supply pressure and as a consequence high back pressure in the discharge piping shall be avoided by proper sizing of orifices, lines and valves as well as operating procedures for manual depressurization of the pipeline.

Overpressure protection against overpressure for drain piping system shall be realized by proper operating procedures for pigging operation. Opening of drain valves while the pig launcher is pressurized shall be prevented to ensure protection of personnel.

4. Isolation Methods and Requirements

Adequate isolation method shall be provided for all process and utility systems to ensure proper operation of the Facilities while some sections/equipment are taken out of the service for maintenance, inspection or hot work. Isolated sections/equipment shall be designed in such a way to ensure safe working environment during maintenance work.

The methods of isolation may be categorized into positive isolation and valved isolation.

4.1. Positive Isolation

Positive isolation shall be required where no leakage can be tolerated for safety and contamination reasons, e.g. multiple stream entry into a common header. Positive isolation shall be achieved by using spectacle blinds (by turning), by insertion of a spade between flanges, replacement of a spacer with a spade or removal spool piece with blanking of open ends. All above mentioned mechanical barriers shall follow the piping specification (material, pressure and temperature rating and facing) in which they are installed.

Positive isolation enables the following:

- To enable the entire system/section to be taken off-line for inspection, maintenance or hot work without interrupting the operation of adjacent units.
- To provide physical proof of safe and total isolation before inspection, maintenance or replacement of single pieces of equipment involving breaking of containment, e.g. control valves.
- To facilitate operation of safe and effective "Permit to Work" system for vessel entry and other confined space entry.
- To prevent utilities system from becoming contaminated with process fluid.
- To enable installations of tie-ins to new equipment.
- Hydro testing of equipment.
- For process fluids at or above auto-ignition temperature.
- Where equipment is to be mothballed.

Positive isolation element shall be located in readily accessible locations, close to the system (within skid or package), to reduce the volume for depressurization, venting and draining. Its location shall enable the operator to monitor and confirm isolation from where the work is to be performed.

Positive isolation elements shall be purchased from approved vendors as engineered items and not locally made. It shall be permanently marked with the manufacturer, model number, size, material

and pressure rating.

The design shall ensure that the amount of flange spread necessary to remove and insert spades, as well as the orientation of the flanges are made available. The design shall also include spacers at difficult essential spading points.

The design shall consider piping support and access requirements where a removable spool or spading point is provided.

4.2. Valved Isolation

Valved isolation shall be used to maintain containment while positive isolation is being made.

Means to confirm the integrity of the isolation prior to breaking containment shall be provided wherever valved isolation is used as the primary protection. This shall be achieved using a single drain/vent point or body bleeder to confirm satisfactory depressurization/drainage.

The bleed valve shall be routed to a safe location which may be to atmosphere if small volumes are involved or to a closed flare/vent/drain system if large volumes are involved.

Where bleed is connected to a closed system, an atmospheric drain/vent tapping should be provided to confirm adequate depressurization before breaking is proceeded.

Valved isolation shall be achieved using reliable manually operated valves capable of tight shutoff. Suitable rated ball, plug or gate valves are considered effective as block valves for isolation.

Control valves shall not be considered to form part of valved isolation.

Globe valves and needle valves shall not be used for the isolation scheme. PSV or Check valves also shall not be used for isolation.

Valved isolation allows flow of fluid to be stopped and started for the following:

- To maintain containment while positive isolation devices are being installed.
- During start up and shutdown of the whole facility or parts thereof.
- For re-routing of fluids within the process as required by changing process conditions or requirements.
- To allow draining, purging, and depressurization.
- Low risk, non-entry situations.
- For short term isolation of systems.

Valved isolation shall be provided as given in Table 1.

Service Temperature and Pressure		Non-toxic and flammable		Non-toxic and non-flammable
		Above auto ignition temp. or flash point < ambient	Below auto ignition temp. or flash point < ambient	
Above 50 barg or above 200°C		V - DBB	V - DBB	V - SVI
Between 2 and 50 barg and below 200°C		V - DBB	V - SVI	V - SVI
Below 2 barg and below 200°C		V - SVI	V - SVI	V - SVI
Key	V =	Initial valving of live system to allow for further isolation to proceed.		
	DBB =	Double block and bleed isolation as defined in paragraph 4.2.1.		
	SVI =	Single block and bleed isolation as defined in paragraph 4.2.2.		

Table 1 Minimum isolation categories for different conditions and service

4.2.1. Double Block and Bleed Isolation

Double block and bleed isolation includes:

- two block valves with an interposing bleed valve.
- single body integral DBB with two balls and bleed valve (monoblock).

The integrity of the single two block valves without intermediate bleed shall only be considered same as single valve – inability to confirm the integrity of the upstream valve.

Shutdown valves may be considered to form part of isolation valving on a case-by-case basis, subject to approval by BFPCL. If a shutdown valve is used, e.g. as one of two block valves, it must be ensured that the integrity of the rest of the shutdown system is not compromised. The valve shall be disabled and mechanically secured/locked closed. Isolation of the primary motive power (electrical, pneumatic or hydraulic) may also be necessary.

Instruments such as pressure tappings shall be provided with isolation facilities as per Table 1. Instrument valves shall not be considered to provide part of a DBB arrangement i.e. DBB valves shall be to piping specification. Integral piping DBB valves shall be preferred for instrument isolation i.e. FLG / NPT single body two valve assembly with bleed between valves.

4.2.2. Single Block and Bleed Isolation

Single Block and Bleed valve isolation includes single isolation valve and bleed valve on side of the

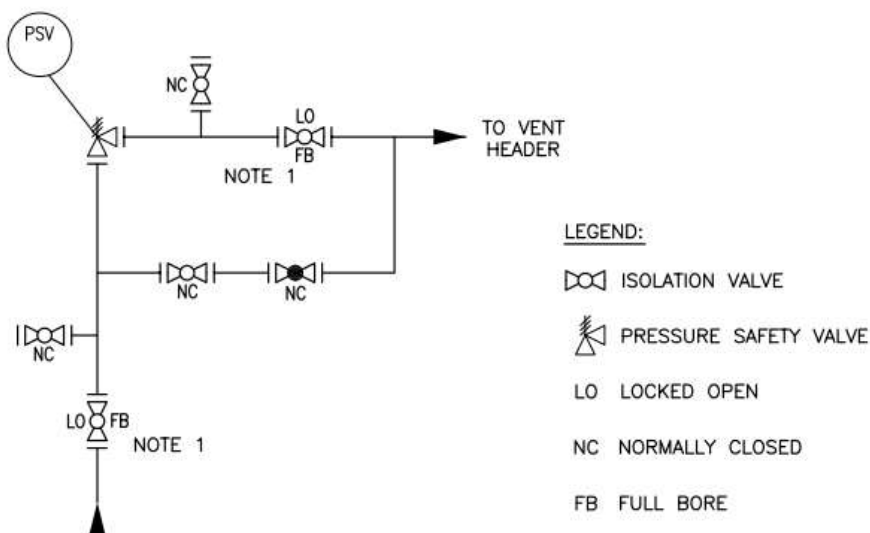
equipment / section to be taken off-line.

4.2.3. Pressure Safety Valves

Pressure safety valves are required to be inspected and tested at regular intervals to ensure safe operation and for critical systems a spare relief shall be provided to keep the system in operation while the PSV is taken out of service for testing or maintenance.

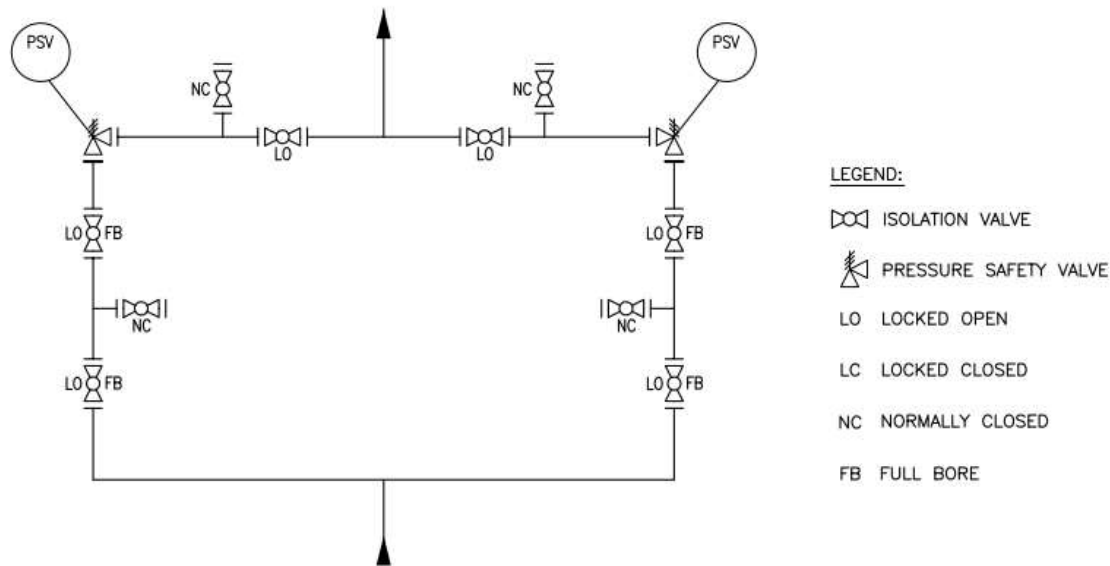
For systems where single relief valve is installed, isolation will not be provided at upstream section of relief valve to ensure that relief valve is not isolated from the system (unless isolation valve is indicated as LO). Single block and bleed isolation shall be provided at downstream of relief valve venting to a vent system. This isolation valve shall be indicated as LO. The valve shall be full bore or in case of reduced bore, the valve port shall have at least the area of the relief valve outlet. Typical arrangement is shown in Figure 1.

For systems where spare relief valve is installed, isolation valve as per isolation category presented in Table 1 shall be provided at upstream section of relief valve to allow the protected system to be kept in operation while the PSV is taken out of service for testing or maintenance. This isolation valve shall be full bore or in case of reduced bore, the valve port shall have at least the area of the relief valve inlet. Isolation valves upstream relief valve shall be indicated as LO or LC if relief valve is under operation or in service, respectively. Single block and bleed isolation shall be provided at downstream of relief valve venting to a vent system. This isolation valve shall be indicated as LO. The valve shall be full bore, or in case of reduced bore, the valve port shall have at least the area of the relief valve outlet. Typical arrangement is presented in Figure 2.



NOTES:

1. REDUCED BORE VALVE IS ACCEPTABLE IF VALVE PORT AREA IS BIGGER THAN RELIEF INLET.

Figure 1 Isolation for single relief valve*Figure 2 Isolation for relief valve with spare*

4.2.4. Shutdown valves

Shutdown valves at BBOU and Elepa shall be provided with isolation valve located downstream of the SDV. A bleed shall be provided between the isolation valve and shutdown valve to allow for valve leakage testing.

4.2.5. Gas metering skid

Inlet and outlet isolation valves for all trains of gas metering skid (including proving by-pass) shall be single body double block and bleed valve and shall be equipped with body cavity local pressure indication. This configuration is to be applied to gas metering skid at BBOU, Elepa and Akabel.

4.2.6. Pig launcher / receiver

Pig launcher and pig receiver shall be isolated from main pipeline by introducing double block and bleed arrangement. Two isolation valves shall be installed on kicker line and downstream the minor barrel with bleed valve between.

4.2.7. Vent header

Before opening the connections to the process vent system, upstream process utility or equipment must be isolated and depressurized.

Isolation of upstream process facilities shall be provided to ensure positive isolation of any shutdown equipment or system from the vent system, depending on the isolation categories for different conditions and service.

4.2.8. Closed drain

Before opening the connections to the process drain system (closed drain), upstream process utility or equipment must be isolated and depressurized.

Isolation of upstream process facilities shall be provided to ensure positive isolation of any shutdown equipment or system from the drain system, depending on the isolation categories for different conditions and service.

4.2.9. Open drain

Before opening the connections to the open drain system, upstream process utility or equipment must be isolated and depressurized.

Isolation of upstream process facilities shall be provided to ensure positive isolation of any shutdown equipment or system from the drain system depending on the isolation categories for different conditions and service. Additionally, isolation valve upstream the pipe outlet shall be LC to avoid possibility of inadvertent valve opening.

4.2.10. Utility headers

Individual off-takers from the instrument air headers shall be isolated with single isolation valves.

4.2.11. Locked and CSO/CSC Valves

Manual valves may have to be locked in open or closed position in order to ensure that they are not inadvertently changed to the incorrect position. The reasons for locking may include prevention of safety hazard, environmental impact, major equipment damage or commercial regulations.

The valve position shall be secured by key locked – LO (Locked Open) / LC (Locked Closed). A system shall be put in place and executed for the effective and safe management of the locked valves.

Locking device shall be used as indicated below:

- Locked open / locked closed isolation valves upstream and downstream PSV with spare (for details see paragraph 4.2.3).

- Locked open isolation valve upstream and downstream single PSV (for details see paragraph 4.2.3).
- Locked closed isolation valve upstream the manual pipeline depressurization valve (restriction orifice).
- Locked open isolation valve upstream the blowdown valve.
- Locked closed isolation valve on bypass line of the blowdown valve.
- Locked closed isolation valves on bypass lines of the metering skid individual runs inlet valves
- Locked open isolation valve at tie-in to the vent header at NGC.
- Locked closed isolation valve upstream pipe end to open drain (quick connection coupling).
- Locked closed isolation valve on bypass line of the outlet isolation valve (upstream and downstream barred tee at GRF and NGC, respectively).
- Locked closed isolation valve on bypass line of the shutdown valve (SDV) upstream barred tee.
- Locked closed isolation valve on bypass line of the isolation valve inlet to metering trains;
- Locked closed isolation valve on inlet line to metering trains from the prover.
- Locked open isolation valves at inlet/outlet to/from metering skid.

Car seals shall be plastic or metal strips installed to prevent changing of the valve position without cutting the seal (it occurs if position of locked valves is being changed).

Car sealing device shall be used as indicated below:

- Car seal open isolation valves on instrumentation with trip functions

Exceptions from above described sealing device philosophy are subject to approval by the BFPCL.

4.2.12. Isolation of Instruments

Instruments such as pressure tappings must be provided with isolation facilities in accordance with the mechanical section of this philosophy. Integral isolation valving is to be considered for field instrumentation when appropriate for that service.

Instrument valves shall not be considered to provide part of a DBB arrangement i.e. DBB valves shall be to piping specification.

Integral piping DBB valves shall be preferred for instrument isolation i.e. FLG / NPT single body two valve assembly with bleed between valves.

All pressure instruments shall be equipped with isolation valves with minimum requirements in accordance with Table 1. Instruments with trip functions connected to the ESD system shall be CSO.

5. Instrumentation and Control System Philosophy

5.1. Process Control

The BBOU, Elepa and Akabel Process Control will be done via a Process Automation System supporting 4-20 mA signal with HART protocol or similar technology that offers same advantage. Safeguarding functions will be separately controlled via a safety instrumented control system (SIS-System) certified to required SIL level.

EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-004 - Control System Block Diagram – BBOU, EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-005 - Control System Block Diagram – ELEM and EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-006 - Control System Block Diagram – AKAB will indicate location of cabinets and control equipment.

Plant instrument signals will be monitored by means of several Human Machine Interface (HMI) Terminals located at each Manifold control room. The remote Manifolds shall be equipped with a remote terminal unit which transfers all process signals via redundant fiber optical data transfer to the main control room at Akabel.

A remote-control unit will be installed at the BBOU and Elepa Manifold Area. Remote I/O unit shall be capable to operate PAS and PMS (non-SIF classified). Most likely two/three separate remote I/O's with communication processor will be installed in a dedicated Manifold cabinet. Communication between Cabinet and Akabel Receiving Facility shall be done via FO-Line.

On loss of power the remote-controlled units shall be continuing to operate via a dedicated UPS. UPS sizing shall be based on generic PMS, PAS and SIS specification. Duration of UPS will be 8 hours minimum.

A separate UPS cabinet (batteries and controller) is to operate each of the Manifold Platform. The separate UPS cabinet expected to be of SS material and mounted inside the control room building.

5.2. Pressure Instruments

5.2.1. General

The measuring of pipeline/ facilities pressure shall be used for

- monitoring of actual pipeline pressure
- low/ high pressure protection
- recording of actual pipeline/ manifold piping pressure
- pressure compensation for flow metering
- actual value signaling for pressure control of station

Intelligent pressure transmitters (SMART type) with Liquid Crystal Display (LCD) are proposed for all pressure values that need remote transmission and pressure gauges for local pressure indication.

The Transmitter power supply shall be 24V DC and shall be HART/ intelligent 4-20mA transmitter on two wire principles (Loop powered).

5.2.2. Pressure Gauge

The gauge body, wetted parts and element material shall be as indicated in the data sheets.

Bourdon type design compliant with operational safety requirements according to DIN EN 837-1 pressure gauges filled with fluid shall be used. Pressure gauge ranges shall be selected; according to the piping pressure classes such that the normal operating pressure is between 40 to 70% of full scale. Gauges shall be able to withstand over-range pressure of 1.3 x calibrated range as standard. Pressure gauges shall be used with 150mm diameter (measured range on over 270 degrees) and stainless steel casing.

- The gauges shall have an accuracy of $\pm 0.5\%$ of full scale reading.
- The Ingress Protection shall be at least IP66 as per IEC 60529.
- The scale of gauges shall be single type.

All gauges shall be full safety pattern fitted with shatterproof safety glass and blow out back protection. Gauges exposed to vibration or process pulsations shall be fitted with pulsation dampers to the same material as the element.

Gauges process connection shall be thread $\frac{1}{2}$ " NPT M.

The Quantity, Range, Material, Tag Number and any additional information shall be defined in the data sheet for the Pressure Gauge mentioned in chapter 3 "Referenced Documents".

5.2.3. Pressure Transmitter & Differential Pressure

The transmitter body, wetted parts and element material shall be as indicated in the data sheets.

The Differential Pressure and Pressure Transmitters shall be working on industrially latest up-to-date proven technology.

The type of transmitters shall be digital type, SMART series transmitters with HART protocol and built-in temperature and static pressure compensation.

HHC (Hand-Held Communicator) shall be supplied by Supplier of the instrumentation for the project (two for each VENDOR).

Over range pressure protection shall be able to protect the sensing element from the maximum hydrostatic test pressure applied to either side, with the opposite side vented to atmosphere. Pressure transmitter body shall be capable of withstanding 150% over range pressure without damage. The transmitter body and measuring element shall have a rating of not less than 3000 psig (206.8 bar).

The transmitter shall be provided with an integral local digital indicator as standard. The integral indicator shall indicate the pressure / differential pressure / flow etc, in engineering units (inclusive of square root extraction, where used).

The transmitter sensor shall be provided with a silicone oil or equivalent fill fluid to withstand the maximum operating pressure and temperatures.

Pressure transmitter instruments shall be provided with a process isolating 2-valve manifold with vent connection to facilitate calibration and depressurization prior to instrument removal according to the approved Datasheets and instrument hook ups.

The impulse pipe work arrangement for differential pressure measurement shall be the same as for two separate pressure measurements. On no account shall a differential pressure gauge be supported by its impulse pipe work. It is important that seal chambers, diaphragm seals, condensate pots or filling tees shall be at the same level and the impulse pipe work shall be run together to avoid static pressure errors.

Differential pressure transmitter instruments shall be provided with a process 5-valve manifold with vent connection to facilitate calibration and depressurization prior to instrument removal.

The transmitter minimum working voltage required at the transmitter terminals shall be clearly defined in the manufacturer's documentation.

Transmitters process connection shall be thread $\frac{1}{2}$ " NPT F.

Response time shall be less than 1 second.

Where there is lots of particle or crystal, a flushing ring shall be packed with transmitter. The connection size of flushing ring shall be $\frac{1}{4}$ NPT (F)".

The minimum specification requirements for the pressure and differential pressure transmitters are as follows:

- Accuracy $\pm 0.075\%$ of the calibrated span (including temperature drift and linearity).
- Hysteresis and dead band 0.2% of calibration span.
- Repeatability ± 0.1 of calibration span.
- Stability / drift rate 0.1 of Upper Range Limit (URL) for the period of 6 months.
- Static pressure effect on DP Delta Pressure (DP) 0.1 of URL per 70 bar or better
- Ambient temperature effect within 0.2% of calibrated span per 28°C .
- Span as per the instrument data sheets.
- Range as per the approved instrument data sheets.
- Hazardous Classification shall at least NEC Class 1, Division 2, Group C&D, T2
- Ingress Protection shall be at least IP66 as per IEC 60529.
- Certification for transmitter to use shall be Class 1, Div 2, Grp C&D, and Temp. Class T3. The preferred method of protection is flameproof EEx'd (ATEX).
- Span/range units pressureBar g
- Span/range unit's differential pressure m bar/MM H₂O.
- Silicon strain gauge capsule and capacitance diaphragm sensing elements are preferred for pressure and differential pressure duties, respectively.
- HART intelligent protocol electronics shall be fitted. Consideration must be made for the effect of high temperatures conducting up to the transmitter electronics. Where the installation requires use of a siphon, it shall preferably be integral with a gauge union / velocity check.
- The body material shall be compatible with the process fluid, with SS 316 stainless steel as a minimum.
- All pressure and differential pressure transmitters shall have local digital indication.
- Differential pressure transmitters shall be capable of withstanding full design pressure on either side without affecting calibration.
- Transmitters shall be used for all differential pressure measurement.

The Quantity, Range, Material, Tag Number and any additional information shall be defined in the data sheet for the Transmitter.

5.2.4. Pressure Switch

The pressure switches' body, wetted parts and element material shall be as indicated in the data sheets.

The pressure switches shall be working on industrially latest up-to-date proven technology.

The pressure switch shall be used in pneumatic applications and shall be from type gauge pressure switch.

The pressure switches process connection shall be threading ¼" NPT F.

All relevant data for the pressure switches as Set point adjustments, process data, maximal pressure ratings, elements, switch, enclosure, etc. shall be according to the approved datasheets for PSYHH and PSYLL switches.

The pressure switches PSYLL and PSYHH shall be used for indication the pressure LL (Low Low) and pressure HH (High High) into the Main Control Room via electrical contacts inside of pressure switches.

The manufacturer instruction in the pressure switch manual for installation, operating and maintenance shall be strictly followed.

The Quantity, Range, Material, Tag Number and any additional information shall be defined in the data sheet for the Pressure Switch.

5.3. Temperature Instruments

5.3.1. General

Apart from special applications for equipment or machines, the temperature elements shall be generally in suitable thermowells.

The measuring of temperature shall be used for

- monitoring of actual product temperature in pipeline
- temperature compensation for flow metering

All temperature elements shall be installed according to Typical Instrumentation hook up.

Temperature measurement shall normally be made by resistance thermometer (RTD), or bimetallic thermometers.

5.3.2. Temperature Gauge

Dial thermometers with quick response time shall be used for local indication.

The Temperature Gauge shall be of bi-metal thermometer type. The same shall be of heavy-duty weatherproof construction. The diameter of the dial shall be 150 mm (measured range on over 270 degrees) and hermetically-sealed in stainless steel case.

The Quantity, Range, Material, Tag Number and any additional information shall be defined in the

data sheet for the Temperature Gauge mentioned in chapter 3 "Referenced Documents". The length and diameter of the stem shall be suitable for the thermo well.

Temperature gauge shall be connected with thermo wells with thread $\frac{1}{2}$ " NPT M.

Tolerance class shall be "Class A" in accordance with IEC 60751, calibration in °C.

Only filled thermal systems shall be applied.

5.3.3. Thermo Wells

Thermo wells shall be solid drilled and tapered.

Thermo wells shall be compliant with IEC 61520.

Thermo wells shall have 1 $\frac{1}{2}$ " flange RF (Raised face (flange)) connections with process and $\frac{1}{2}$ " NPT F thread for temperature elements. Thermo well pressure class shall be defined in datasheet.

The well and flange material shall be suitable for process medium, design and piping class.

Thermo wells shall be installed into the pipe in that way, that end of thermo well is placed in the range from $\frac{1}{3}$ ID to $\frac{1}{2}$ ID.

Thermo wells shall be installed that the element and the head can be removed without bending the element or dismounting other equipment.

Thermowell shall be machined from bar stock material. Other materials shall be furnished as required in the piping specification.

Calculations shall be tabulated by the thermo well supplier in order to show that all thermo wells are suitable for stress due to maximum stream velocity conditions (Vortex shedding effect- In fluid dynamics, vortex shedding is an oscillating flow that takes place when a fluid such as air, gas or water flows past a cylindrical body at certain velocities, depending on the size and shape of the body).

Thermo well wake frequency calculations shall be carried out in accordance with ASME PTC 19.3.

5.3.4. Temperature Elements/Transmitter

The resistance thermometer (RTD) elements shall be of the platinum type Pt (100 ohm at 0°C), fundamental interval 38.5 ohms in accordance with IEC 60751 tolerance Class A.

Pt100 RTDs may be used over a practical range of -200°C to +400°C; when the temperature is above 400°C, thermocouple shall be preferred.

Temperature element stem shall be AISI 316L stainless steel as minimum. Suitable thermal paste shall be filled into the thermo well before insertion of the temperature element. Preferred are spring loaded temperature stems.

For general purpose applications a 3 wire RTD configuration shall normally be used.

Temperature transmitters shall be connected with thermo wells with thread ½" NPT M.

The type of transmitters shall be digital type, SMART series transmitters with HART protocol and in advance diagnostic and alarm ability. The output signal shall be 4 to 20mA DC. Loop powered 2-wire signal transmission shall be used.

The transmitters shall be installed as close as possible to the sensor, preferably in the sensors head. As a minimum the transmitter shall provide cable lengths compensation and linearization of the Pt100 sensor characteristic.

All temperature transmitters shall have a galvanic separation between the sensor element and the output amplifier. The transmitter shall have sensor open circuit detection feature.

Response time shall be less than 1 second.

The accuracy shall be $\pm 0.2\%$ reference accuracy of transmitter characteristic curve nonlinearity.

The minimal specification requirements for the temperature transmitters are as follows:

- D/A (digital-to-analogue) Accuracy ± 0.05 of calibrated span (including temperature drift and linearity).
- Cold junction accuracy ± 0.5 Deg C.
- Hysteresis and dead band 0.2% of calibration span.
- Repeatability $\pm 0.1\%$ of calibrated range.
- Span as per the instrument data sheets.
- Range as per the approved instrument data sheets.

The transmitters shall be suitable for mounting in any position without affecting their measurement accuracy or linearity.

The transmitter shall have an in-built selectable upscale/downscale burn out protection. The transmitter shall be provided with an integral LCD as standard and shall indicate the measurement in °C.

The Quantity, Range, Material, Tag Number and any additional information shall be defined in the data sheet for the Temperature Transmitter.

5.4. Shutdown Valve

Shutdown valve (SDV) shall be full or reduced bore valve according to the datasheet. The SDV shall be supplied complete with actuator and accessories as filter, filter regulator, limit switches, solenoid valve, pilot valve, pressure indicator, pressure safety valve according to the approved datasheet of SDV.

Valve stems shall be fitted with an indicator to determine the position of the valve irrespective of the actuator position. This is to ensure the true position in case of actuator failure/ shearing of the valve stem.

The nominal size of valve shall be in accordance with the approved datasheet.

All shutdown valves shall be fire safe to API 607 or 6FA or equivalent fire testing standards.

The SDV shall be gas over oil, failure-close, normally energised.

The process connection of the SDV shall be flanged in-line into the pipeline, and the ANSI flange size and rating shall be as specified in the datasheet. Flanges shall be in accordance with ASME B16.5 or B16.47 as applicable.

Flange face surface finish shall be Raised Face in accordance with ASME B16.5.

Valve face-to-face dimensions shall be in accordance with API 6D.

The material of the SDV shall be in accordance with the requirements of the piping class and identified in the datasheet.

Shutoff valve seat leakage shall be tight shut-off, class VI, in accordance with IEC 60534-4.

Where soft (resilient) inserts are required for meeting the specified leakage rate, the inserts should be of glass-fibre-filled or graphite-filled Poly Tetra Fluor Ethylen (PTFE). The selection shall be based on the suitability for the specified process.

The resilient material shall be properly clamped between the metal parts and/or locked in position to prevent blowout in the closed position.

The SDV shall be supplied with permanently stainless steel plate, which contained at least the following information:

- Tag number
- PO number
- Manufacturer's name or trade mark
- Manufacturer's model/type number (valve and actuator)
- Manufacturer's serial number
- Body pressure rating
- Size (body and trim)
- Material (body and trim)
- Action on air supply and/or signal failure

Valve body flanges, etc. shall be marked in accordance with the applicable design code.

The direction of flow, the valve body rating, the body material and Tie Shut Off (TSO- means Zero/minimum leakage shall be expected for SDV. Generally a SDV seat leakage shall pass the seat leakage testing per API 508 and/or ISO 5208) direction shall be clearly indicated by permanent mark cast or stamped on the valve body, not painted.

The valve tag number shall be stamped on the valve body, if the identification plate is attached to the actuator.

The Quantity, Range, Material, Tag Number and any additional information shall be defined in the data sheet for the Shutdown valve mentioned in chapter 3 "Referenced Documents".

5.5. Pig Signaller

5.5.1. Intrusive Pig Signaller

The intrusive pig signaller shall be performed with a mechanical contact. In case of passing of a pig, the signaller shall provide visible signalisation via mounted mechanical flag. The pig signaller shall perform with a manual RESET. The pig signaller shall be mounted with a 2" flange.

The intrusive pig signaller shall be connected to the pipeline, to the pig launcher and pig receiver using 2" Flange.

The pig signaller body, wetted parts and element material shall be as indicated in the approved data sheets.

An Ball Valve shall be installed according to the Hook up in order to have the possibility to disconnect in case of failure from the pipeline.

The Pig Signaller shall be working on industrially latest up-to-date proven technology.

The manufacturer instruction in the pig signaller manual for installation, operating and maintenance shall be strictly follows.

The Quantity, Range, Material, Tag Number and any additional information shall be defined in the data sheet for the Pig Signaller mentioned in chapter 3 "Referenced Documents".

The intrusive Pig signaller shall be by manufacturer Online Electronics Ltd or similar.

5.5.2. Non-Intrusive Pig Signaller

The non-intrusive magnetic pig signaller shall detect pigs which are fitted with rare earth magnets. In case of passing of a pig, this event shall be signalled via a graphical display local.

The non-intrusive pig unit shall be attached to the pipeline, to the pig launcher and pig receiver using ratchet straps or stainless steel banding.

The pig signaller body, wetted parts and element material shall be as indicated in the approved data sheets.

The Pig Signaller shall be working on industrially latest up-to-date proven technology.

The manufacturer instruction in the pig signaller manual for installation, operating and maintenance shall be strictly follows.

The Quantity, Range, Material, Tag Number and any additional information shall be defined in the data sheet for the Pig Signaller mentioned in chapter 3 "Referenced Documents".

The non-intrusive Pig signaller shall be by manufacturer Online Electronics Ltd or similar and the type shall be 4001D MAGSIC or similar.

5.6. Restriction Orifice

Restriction Orifice is used in the blow down line to the Vent Header.

The restriction orifice shall be as indicated in the data sheet.

Orifice plates shall be according to process design (temperature, pressure), process medium, and mounted between flange tap type orifices flanges complete with jack screws.

Orifice plate design and dimensions shall conform to the recommendation of ISO 5167. Surface roughness shall not exceed the criteria set in ISO 5167.

All restriction orifices shall be sized not to exceed a 0.60 beta ratio according to DPR standard "Procedure Guide for the Determination of Quantity and Quality of Gas at Production and Custody Transfer Points/ chapter 6.5.4 Meter runs and header design for orifice meters". Pipe shall conform to material and schedules called for in the Piping Specifications.

The Quantity, Range, Material, Tag Number and any additional information shall be defined in the data sheet for the Restriction Orifice.

5.7. Additional Instrument and Equipment types

5.7.1. Limit Switches

Shutdown valves shall be fitted with limit switches which indicate fully open and fully closed positions.

The limit switches shall be inductive proximity switches NAMUR type, two wires in accordance with DIN 19234 and VDI/VDE 3845. Position switches have to be designed with intrinsically safe standard for hazardous area approval. Limit switches for hand operated valves shall be installed in such a manner that accidental damage of the switches is avoided. The limit switches used for ESD application shall be as per the approved certified SIL.

The proximity switch shall have a flying lead termination in an enclosure mounted close to the switch. The proximity switch and enclosure shall be rated to IP65. In case of more than one proximity switch all shall be terminated in one enclosure with sufficient cable entries. The flying lead shall be protected by stainless steel helical shroud.

The local enclosures shall be made from stainless steel. Cable entries to the enclosure shall be via isometric threaded entries ISO M20x1.5mm.

Proximity switches shall be positioned to have free maintenance access and avoid accidental damage and vibration.

Mechanical action switches shall be hermetically sealed and rated to IP65 as a minimum.

Switch contacts shall be rated at least to 24 VDC, 1A, made from noble metal. All electrical contacts shall be of type SPDT Proximity.

The Quantity, Range, Material, Tag Number and any additional information shall be defined in the data sheet for the Shutdown Valve.

5.7.2. Solenoid valves

Solenoid valves shall be normally used in pneumatic or hydraulic power lines, acting as the actuating device, operated from a control system, to cause full opening or closing of pneumatically or hydraulically powered valve.

The solenoid valve shall be suitable for 24 VDC supply on continuous duty (low power solution), explosion-proof (Exd) according to IEC 60079 and at least IP 65 enclosure.

Solenoid valves shall be suitable for automatic reset, except where process owner indicates manual reset.

Solenoid valves shall be 3 ways universal direct acting type (spring return type). The coil and spring shall be selected to provide sufficient operational power to position the valve stem correctly under service conditions. Coils shall be rated for continuous operation. Coils shall include fuses and override diode and shall be replaceable.

Solenoid valves shall be energised in normal running conditions and de-energised to trip.

Solenoid valves shall be supplied with class H insulation

Coil insulation shall be Class A to IEC-60085 as a minimum. Integral terminals shall be provided for the termination of cabling. Flying leads extending from the coil enclosure shall not be used. Cable entries shall be via isometrics threaded entries ISO M20 x 1.5mm.

The body and trim material shall be according to process design and process medium. The seat and seal material of the solenoid valve shall be as indicated within the approved data sheet of shutdown valve. All wetted parts material shall be compatible with the specified process fluid.

Fail-safe position shall be considered in designing the solenoid valve (electrical failure).

Solenoid valves used for ESD application shall be as per the approved certified SIL.

Three port solenoid valves shall be used for single acting spring return actuators and five port solenoids shall be used for double acting actuators.

Solenoid valves shall be able to operate at the maximum instrument gas/air design pressure (120 psig) in the plant. An instrument air skid for the individual manifold will be responsible for ensuring the proper operations of the pneumatic systems.

Solenoid valves shall be approved to the applicable hazardous area.

The Quantity, Range, Material, Tag Number and any additional information shall be defined in the data sheet for the Shutdown Valve mentioned in chapter 3 "Referenced Documents".

The manufacturer instruction in the solenoid valve manual for installation, operating and maintenance shall be strictly followed.

6. Electrical Installation Requirements

6.1. General

All junction boxes, cables, cores and terminals shall be identified in accordance with project requirements.

Only one core shall be used in each termination. Common connections shall be made using the terminal manufacturer's standard bridging arrangement.

All spare cores to be terminated at junction box terminals and identified as spare. All spare entries shall be sealed by a method that ensures integrity of the junction box. Entries shall be M 20 x 1.5 mm where practical.

Cable glands on outdoor service shall be shrouded.

Cables shall enter and exit junction boxes via the bottom or side to minimise the possibility of water ingress.

Redundant data highway cables for monitoring and control systems, dual UPS supplies and other designated high integrity circuits shall be run by separate routes.

The minimum spacing between instrument and power cables shall be according to acceptable industry standards. These spacing may be relaxed at entries to instruments and associated housings or where, due to physical constraints, these spacing cannot practically be achieved. All such instances must be highlighted in the design.

Special consideration shall be given to the method of cable interconnection across module boundaries. Cable transits shall be specified where possible.

6.2. Earthing System for Instrumentation

Earthing and Lightning shall be performed in accordance with EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-029, EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-030, EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-031, EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-032, EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-033, EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-034.

6.3. Junction Boxes

Instrument IS and hazardous area junction boxes shall be fabricated according to the environmental conditions. Junction boxes for use in hazardous area must be certified for hazardous area zone 1, with a degree of protection of IP65. Junction boxes for use with intrinsically safe circuits must be

dust- and waterproof and have a degree of protection of IP65.

Junction boxes for instrument cables shall be installed at the locations indicated on the layout drawings in such a way that they are permanently accessible.

The junction box shall be supplied with bottom entry gland/ continuity plates.

All cables shall enter via bottom plate.

Sufficient terminals shall be installed in junction box for terminating all cables pairs/cores including spares.

Junction boxes shall be used for connection of single control, signal and instrumentation cables with summarized common cables and for connection of different local equipments.

Different signal types shall not be mixed in the same junction box and separate junction boxes shall be provided for each of the following groups of cables:

- Cables carrying non-intrinsically safe digital signals
- Cables carrying intrinsically safe digital signals
- Cables carrying non-intrinsically safe analogue signals
- Cables carrying intrinsically safe analogue signals.

The junction box number shall be clearly indicated by permanently attached label.

Junction boxes for indoor and outdoor use shall be preferably of the same construction. For indoor application **and** outdoor generally stainless steel shall be used.

Junction boxes to be installed in hazardous areas shall be certified EExe or EExi for intrinsically safe circuit. Junction boxes material shall be stainless steel. Junction boxes shall be equipped with the earth bar for shield connection. Metallic junction boxes shall be equipped with earth bolt. Junction boxes used for intrinsically safe circuit shall be identified with a blue label and provided with blue terminal blocks.

The supports and auxiliary structures for junction boxes shall be made of hot dip galvanized steel. Drilling holes for cable entry have to be located on the bottom side of the boxes; only threaded cable glands shall be used.

For the different cable types (power cables, control, signal and instrumentation cables) and also for different voltage levels separate junction boxes shall be used. The junction boxes shall be tagged according to their application. Color coding shall be used to differentiate all signal junction boxes

from power junction boxes.

Entries shall be M 20 x 1.5 mm where practical. Multi-pair cable entries shall be ISO M25, M32, or M50 as applicable.

To facilitate the termination of any cable screens a bus bar shall be installed as close to the terminal rail as is practical. It shall be mounted on insulated supports and supplied clamps.

Cable carrying different characteristic signals shall be segregated in cable runs and be connected to dedicated junction boxes, for example on/off switch signals and 4 - 20 mA signals.

All junction boxes shall be installed to allow adequate room for the loops of bottom entry off-skid cables. All junction boxes shall be provided with glands and shrouds for the off-skid cables. All junction boxes shall be provided with glands and shrouds for the incoming /outgoing cables.

Mixing the signals for different systems in one Junction Box (JB) shall be not allowed. Separate JBs shall be used for different type of signals e.g. Analogue (4-20mA DC) and 24 VDC, IS and NON-IS, etc.

All terminals for I.S. circuits shall be blue colored.

For junction boxes on shutdown service the background color shall be red and a further label shall be fitted stating "This box contains trip circuit".

For intrinsically safe junction boxes the background color shall be blue and further label shall be fitted stating "This box contains I.S. circuit".

6.4. Instrument Grounding

Field instruments bodies, junction box housings, Local Control Panels housings and any other system cabinet shall be solidly grounded to the plant ground according to EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-029 - Instrument Earthing/ Grounding Details.

The cable armour shall be earthed at the plant end to an electrical earth boss welded to the structural steelwork. Screening shall be earthed at the control panel end only, to an instrument copper earth bar insulated from the steel structure.

7. ESD and F&G Systems Philosophy

7.1. ESD System

Emergency Shutdown Systems BBOU, Elepa and Akabel shall operate independently e.g. if BBOIU ESD system is activated, no action is being taken by Elepa ESD system and vice versa.

Following Pipeline shutdown scenarios will apply:

- Automatic blowdown of the Manifold facilities located upstream of each pipeline inlet SDV valve in case of Manifold shutdown with automatic depressurization (EBD - Level 0). All causes shall be inherited from shutdown philosophy.
- The pipeline inlet shutdown valve shall be closed in case of Manifold shutdown with automatic depressurization (EBD - Level 0) and Manifold shutdown without automatic depressurization (ESD Level 1). All causes shall be inherited from existing shutdown philosophy (for details see paragraph 7.1.2).
- Process shutdown of the pipeline connection from BBOU to Elepa and ELEpa to Akabel shall be realized as shutdown level PSD Level 1 (process shutdown that will shut down inlet SDV valve to the pipelines connection) to avoid unnecessary BBOU and Elepa full Manifold shutdown.
- Depressurization of the pipeline facilities installed downstream the pipeline inlet SDV and the pipeline itself will be depressurized manually only, to the cold vent stack system at each Manifold.

7.1.1. Manifold shutdown with automatic depressurization (EBD - Level 0)

In case EBD Level 0 is activated, all SD valves shall be closed BD valves shall be opened. EBD (Level 0) shall be initiated on the following causes:

- Gas detection signal from F&G system.
- Flame detection signal from fusible plugs.
- Manual EBD push-buttons.

Details are given in Cause & Effect Diagram, doc. no EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-007, EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-008, EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-009.

7.1.2. Manifold shutdown without automatic depressurization (ESD Level 1)

In case ESD Level 1 is activated, pipeline inlet SD valves shall be closed. ESD Level 1 shall be initiated on the following causes:

- High-High and Low-Low level from SPDC at custody transfer point.

- High-High level in Drain Drum.
- Low-Low Instrument Air pressure.
- Manual ESD push-buttons.

7.1.3. Process Manifold shutdown (PSD Level 1)

Process Manifold shutdown PSD Level 1 shall be initiated on Low-Low and High-High inlet pressure sensed on the pipeline feed from OPUG, BBOU, NUNR and SEIB at the Manifolds by closing SDV valve inlet from this pipeline. Remaining SDV valves shall be in open positions to allow uninterrupted gas delivery to Akabel GPP.

7.2. Additional requirements for ESD system

7.2.1. Resetting philosophy

The following existing resetting philosophy shall be applied, unless existing resetting philosophy states otherwise:

- Once tripped, neither SDV valves nor power generators move from safe state automatically. Manual reset shall be provided for the power generator. Software reset shall be used for SDV valves.
- ESD signal from the F&G system remains active, until local reset button is pressed on main F&G control unit, to ensure that personnel properly verify cause of the alarm on site.
- ESD tripping signal from mushroom-type push button remains active until the button is pulled back.
- HH pressure and LL pressure tripping signals are active, as long as process parameter exceeds set-point for a trip. As soon as the parameter gets back within design limits, tripping signal is deactivated.
- ESD logic cannot be reset, while any of shutdown causes is active.
- If no shutdown cause is active, operator can reset C&E logic by means of password protected HMI switch. Software resetting the shutdown logic does not open the shutdown valves, it only enables the Open / Start Hand switch provided on the HMI panel of the valve.

7.2.2. Failure mode

All ESD input and output devices shall be designed as fail-safe systems, failing to a de-energized, fail-safe state which shall be shown in the P&ID for each device.

SDV valves shall fail-close on loss of gas, oil or electric power supply. The solenoid valves actuating SDV valves shall de-energized to close the SDV valve.

7.2.3. Logging/recording

The change of state of ESD input devices, by-pass switches, shut-down push/pull buttons, manual resets or equivalent analogue/digital shutdown limits shall be time-tagged and discriminated by device tag number by the PLC Alarm and Events record. This shall utilize a "first-out" system to identify which ESD device tripped first to cause a shutdown.

7.3. F&G System

The primary objective of the F&G detection system is to provide an early warning to personnel of potentially dangerous situations in terms of fire & flammable gas release to the atmosphere.

The objectives of the F&G detection system are:

- To provide early detection of flammable gas release or a fire.
- To communicate this information to personnel permitting the initiation of a response e.g. raising an alarm, evacuation of personnel etc.
- To initiate automatic executive actions such as alarms, extinguishing discharge and the appropriate level of emergency shutdown etc. to minimize the likelihood of escalation.

The F&G system shall continuously monitor the facilities and automatically initiate executive action through the safeguarding systems (ESD) for isolation of the relevant process equipment. Upon receipt of confirmed fire or gas detection within the facilities, the F&G system shall initiate the following through signals to appropriate F&G panels:

- Alert personnel through general F&G alarm.
- Isolate power supply to all electrical equipment.
- Initiate the appropriate level of Emergency Shutdown (ESD).
- Activate automatic fire and gas extinguishing system (if any) and close fire dampers.

7.3.1. Gas Detectors

Flammable gas detectors shall be used to monitor areas of the facilities in which flammable gas accumulate due to accidental release.

Flammable gas detectors shall be installed/arranged such that accumulation of flammable hydrocarbon gases can be detected at the earliest possible stage which can avoid fire upon finding an ignition source.

Flammable gas detectors may also be positioned to monitor the boundary of fire areas, particularly between hazardous and non-hazardous areas. Detector locations shall also be based upon the

relative buoyancy of the principal gas or vapour to be detected.

Open Path Gas detector (Line of sight - LOS) shall be provided for all the Manifolds.

LOS gas detectors (Open path flammable gas detectors) are ideally suited for detecting gas accumulations or gas cloud migration in open facility. LOS gas detectors shall be used to monitor a complete area or to monitor the perimeter of a process unit to detect drifting clouds.

7.3.2. Gas Detection Levels

There shall be two gas alarm settings, a Low Level and High Level for 1.0 LELm and 2.0 LELm of detected gas content, respectively.